MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8295A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, external anode, ceramic-metal

See figure 1

Mounting position: Any

Weight: 3 pounds (1.36 kg) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	F MHz	Ef Vac	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ec3 Vdc	Ib mAdc
Class AB1 (rf or audio)							
Maximum:	30	6.3	3,000	---	600	100	800
Minimum:	---	5.7	---	---	---	---	---
TEST CONDITIONS:	---	6.0	2,500	Adj	500	0	400

ABSOLUTE RATINGS:

Parameter: / Unit:	Ic1 mAdc	Pp W	Pg1 W	Pg2 W	tk sec	Anode core and seal temp °C	Cooling See note 1
Class AB1 (rf or audio)							
Maximum:	0	1,000	0	30	---	250	---
Minimum:	---	---	---	---	180	---	---
TEST CONDITIONS:	---	---	---	---	300	---	See note 2

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Pulsing emission (sinusoid)	-	prr = 10 (min); eb = ec2 = ec1 = 1,000 v	0.65	II	is	70	---	a
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-85.0	-115.0	Vdc
1266	Total grid current	5		0.65	II	Ic1	---	-50	µAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	---	15	mAdc
	Quality conformance inspection, part 2								
1301	Heater current	-		---	---	If	7.7	8.7	Aac
1331	Direct-interelectrode capacitance (ground cathode connection)	-		---	---	Cgp Cin Cout	--- 36.0 16.5	0.12 44.0 20.5	pF pF pF
1316	Amplification factor	-	g1 to g2 amplification; Ec1/Ic2 = 60 mAdc; Eb = 0	---	---	Mu	3.0	3.8	---
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
---	Low-frequency vibration end points:	-							
1261	Electrode voltage (grid)	-		---	---	Ec1	-85.0	-115.0	Vdc
1266	Total grid current	-		---	---	Ic1	---	-50	µAdc
1372	Current division (short pulse, method B)	-	Eb = Ec2 = 500 Vdc; Ec1 = -200 Vdc; egk/ib = 2.5 a	---	---	egk ic2	--- ---	0 400	v ma
	Quality conformance inspection, part 3								
---	Stability	3	Eb = 3,000 Vdc; Ec1/Ibo = 220 mAdc at t = 0; Read ΔIb at t = 180 sec	---	---	ΔIb t	---	22	mAdc
2204	Linear amplifier power output and distortion	3	Class AB1 amplifier; Eb = 3,000 Vdc; Ec1/Ibo = 220 mAdc; Eg1/Ib = 800 mAdc (1-tone); Rk = 2,000 ± 100 ohms; Rg = 1,000 ohms (max); anode tank Q = 10 to 15	--- ---	--- ---	Po 3rd IM 5th IM	1,350 -20 -30	--- --- ---	W dB dB
---	Life test	4	Group C; linear amplifier power output and distortion except, Eg1/Po = 1,400 W (min) (1-tone); t = 500 hours	---	---	---	---	---	---
	Life-test end points:								
2204	Linear amplifier power output and distortion	-		---	---	Po 3rd IM 5th IM	1,250 -19 -29	--- --- ---	W dB dB

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NOTES:

1. With an anode dissipation of 1,000 watts and an incoming air temperature of 50°C at sea level, a minimum of 24 cfm of air shall be passed through the anode cooler. In all applications, an air-system socket, such as the EIMAC SK-184 or SK-184A, or equal, should be used. With an airflow of 24 cfm, the static pressure drop for the tube and socket is approximately 0.13 inch of water. Cooling air should be applied before or simultaneously with the application of electrode voltages, including the heater, and may be removed simultaneously with them. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

2. During all electrical tests involving application of heater voltage, the use of an air-system socket is permissible, with forced-air cooling of 24 cfm maximum for air at 25°C at sea level. Correction may be made for higher altitude or temperature.

(D) 3. The following tests, listed under quality conformance inspection part 3, shall be performed every 3 months, with sample size and failure allowance as follows:

$n_1 = 4$ $c_1 = 0$
$n_2 = 4$ $c_2 = 1$; where c_2 represents the total allowable failures for the first and second samples combined.

In case of a failure on a stability-test sample, the lot shall be reprocessed and a new sample submitted. The linear amplifier power output and distortion test shall be performed on the sample that is used for the stability test. In case of a failure of a linear amplifier power output and distortion test sample, this test shall become a quality conformance inspection part 2 test, with an AQL of 6.5 percent, inspection level of S3. After three consecutive successful submissions, the test may revert to the quality conformance inspection, part 3 test.

4. This is a destructive test. All voltage and current meters shall be calibrated for ±2 percent or better accuracy as installed in the equipment. Prior to the performance of the life test, the tube shall have met the requirements of the linear amplifier power output and distortion test.

5. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

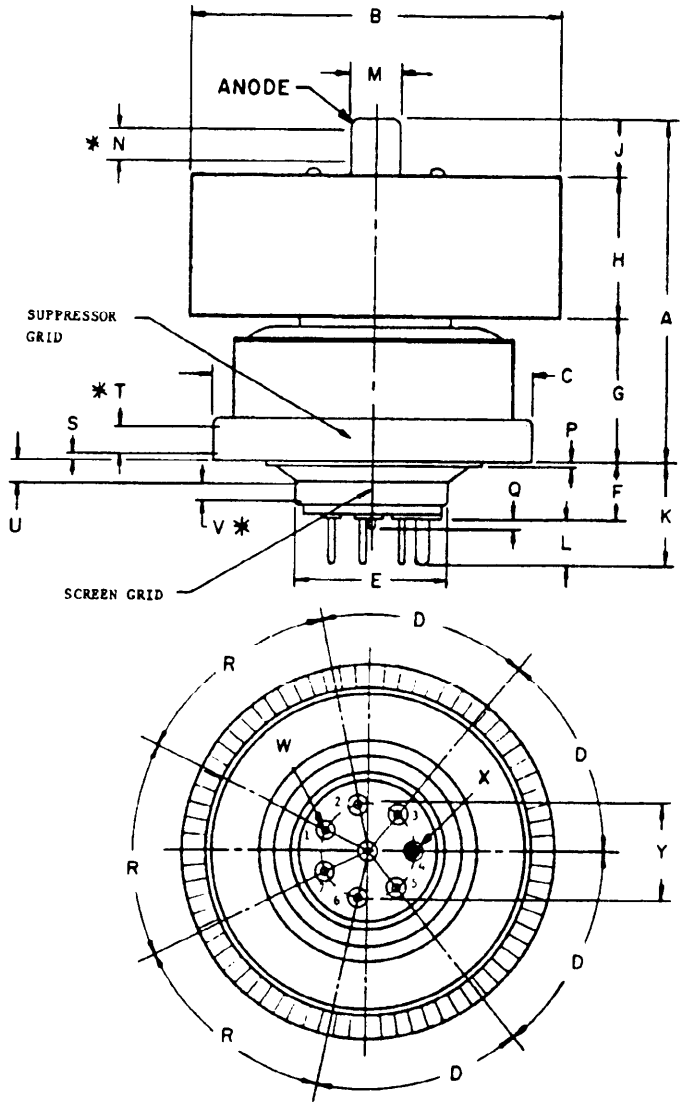
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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* Contact surface shall be confined to this area.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	3.458 (87.83)	3.832 (97.33)
C	3.485 (88.52)	3.515 (89.28)
E	1.615 (41.02)	1.630 (41.40)
F	.655 (16.64)	.719 (18.26)
K	1.056 (26.82)	1.219 (30.96)
M	.559 (14.20)	.573 (14.55)
W	.056 (1.42)	.062 (1.57)
X	.120 (3.05)	.127 (3.23)
Quality conformance inspection, part 3 (see note)		
B	3.968 (100.79)	4.032 (102.41)
G	1.395 (35.43)	1.645 (41.78)
H	1.468 (37.29)	1.532 (38.91)
J	.593 (15.06)	.657 (16.69)
L	.438 (11.13)	.562 (14.27)
P		.125 (3.18)
T	.250 (6.35)	
V	.220 (5.59)	
N	.400 (10.16)	
Reference dimensions		
S	.125 (3.18)	
U	.250 (6.35)	
Q	.125 (3.18)	
Y	1.000 (25.40)	
R	52°	
D	51°	

PIN CONNECTIONS	
Pin No.	Element
1	k
2	g1
3	h
4	k
5	h
6	g1
7	k
Center pin	int con
Lower ring	g2
Upper ring	g3
Cap	a

NOTE: Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL of 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8295A.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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